SUPPLY CHAIN VISIBILITY SYSTEM

WITH OPTIMIZATION ANALYTICS

POWER BI PROJECT DOCUMENTATION

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1. INTRODUCTION

This project presents a Power BI-based Supply Chain Visibility System with Optimization Analytics. The report documents the existing Power BI work across data modelling, KPI creation, inventory analysis, delivery and shipment analysis, supplier scorecards, warehouse efficiency reporting, and executive-level dashboard views.

The solution is designed to bring supply-chain and logistics information into an interactive reporting environment where users can examine orders, order items, customers, products, inventory, shipments, suppliers, cities, categories, and related operational information available in the Power BI model.

The dashboards provide visibility into inventory monitoring, delivery status, supplier-related information, warehouse reporting, revenue, order activity, and selected KPI measures. All descriptions and observations in this document are restricted to the fields, visuals, measures, values, and screenshots visible in the supplied Power BI project.

2. PROJECT OBJECTIVE

- Build a centralized Power BI analytics solution for supply-chain and logistics information.
- Analyze the logistics and e-commerce data available in the Power BI data model.
- Create meaningful KPIs and DAX measures for operational reporting.
- Monitor inventory performance and inventory turnover.
- Analyze delivery and shipment performance, including shipment-status views.
- Evaluate supplier-related information through a supplier scorecard.
- Analyze transportation and shipment activity where the report provides the relevant fields and visuals.
- Monitor warehouse efficiency using the existing Warehouse Efficiency Report.
- Provide executive-level visibility through the Executive Overview dashboard.
- Improve dashboard usability, layout consistency, filtering, and report readability.

3. TOOLS AND TECHNOLOGIES

Technology / Component	Use documented in the project
Microsoft Power BI Desktop	Primary environment used to build the Power BI report and dashboards.
Power Query	Used as the Power BI data-preparation/query layer where applicable in the project.
DAX	Used for KPI and measure creation where measures are present in the report.
Power BI Data Model	Used to organize the project tables and their relationships.
Power BI Visualizations	Used for KPI cards, charts, slicers, tables, and dashboard reporting.

Exact transformation steps, relationship definitions, and DAX formulas that are not visible in the supplied screenshots are documented as “To be updated from the Power BI report.”

MILESTONE – 1

Data Modelling & KPI Foundation

Milestone 1 establishes the data and KPI foundation for the Supply Chain Visibility System. The supplied Power BI screenshot shows the main report page with KPI cards, shipment-status analysis, monthly orders, category-wise orders, city-wise orders, monthly revenue, and an Inventory Turnover (Proxy) visual.

Milestone 1 Deliverables

- Import and prepare the logistics and e-commerce datasets used by the Power BI report.
- Design and use the Power BI data model.
- Create core KPIs and DAX measures present in the report.
- Build the KPI Foundation dashboard and inventory-turnover view.

4.1 Milestone 1 Introduction

The purpose of Milestone 1 is to establish a reliable reporting foundation. It brings the available tables into the Power BI model, prepares the data for analysis, and provides the first set of KPI and operational visuals used across later dashboard pages.

4.2 Data Import and Cleaning

The Power BI report contains multiple tables visible in the Data pane. The supplied screenshot shows the following tables: categories, customers, employees, order_items, orders, payments, products, promotions, returns, shipments, stores, and suppliers.

The screenshots confirm the presence of a Power BI data model and report visuals. Detailed Power Query transformation steps, exact data-type changes, missing-value treatment, and column-by-column cleaning rules are not visible in the supplied screenshots; therefore, those details are not inferred.

- Data import: tables visible in the Power BI Data pane are part of the existing model.
- Data types: To be updated from the Power BI report.
- Missing or blank values: To be updated from the Power BI report.
- Data transformation steps: To be updated from the Power BI report.
- Column preparation: To be updated from the Power BI report.
- Relationship preparation: To be updated from the Power BI report.

4.3 Data Modelling

The supplied Power BI screenshots show the model's table names in the Data pane. The visible tables are categories, customers, employees, order_items, orders, payments, products, promotions, returns, shipments, stores, and suppliers.

The exact relationship lines, cardinalities, cross-filter directions, primary/foreign-key mapping, and the formal fact/dimension classification are not visible in the supplied screenshots. These details are therefore recorded as: “To be updated from the Power BI report.”

Visible table	Status from supplied screenshot
categories	Present in Power BI model
customers	Present in Power BI model
employees	Present in Power BI model
order_items	Present in Power BI model
orders	Present in Power BI model
payments	Present in Power BI model
products	Present in Power BI model
promotions	Present in Power BI model
returns	Present in Power BI model
shipments	Present in Power BI model
stores	Present in Power BI model
suppliers	Present in Power BI model

Data Model screenshot: To be updated from the Power BI report. No separate relationship-model screenshot was supplied with the six screenshots used for this documentation.

4.4 KPI Creation

KPI / Measure visible in report	Visible value in supplied screenshot
Total Order Items	600K
Total Customers	50K
Total Orders	300K
Total Revenue	4bn
AOV	12,759.15
Inventory Turnover (Proxy)	150.05

The exact DAX expressions for these measures are not visible in the supplied screenshots. Exact formulas: “To be updated from the Power BI report.”

4.5 Milestone 1 Dashboard

The supplied Milestone 1 screenshot shows a dark-themed Power BI dashboard titled “Supply Chain Visibility and Inventory Dashboard.” The page contains KPI cards for Total Order Items, Total Customers, Total Orders, Total Revenue, and AOV. It also includes an Inventory Turnover (Proxy) visual, Shipment Status Analysis, Monthly Order, Category-wise Orders, City-wise Orders, and Monthly Revenue.

The visible shipment-status donut displays a total of 300K, with shipped, delivered, and late categories shown in the visual. The category and city visuals provide grouped order views, while the monthly visuals show time-based order and revenue patterns.

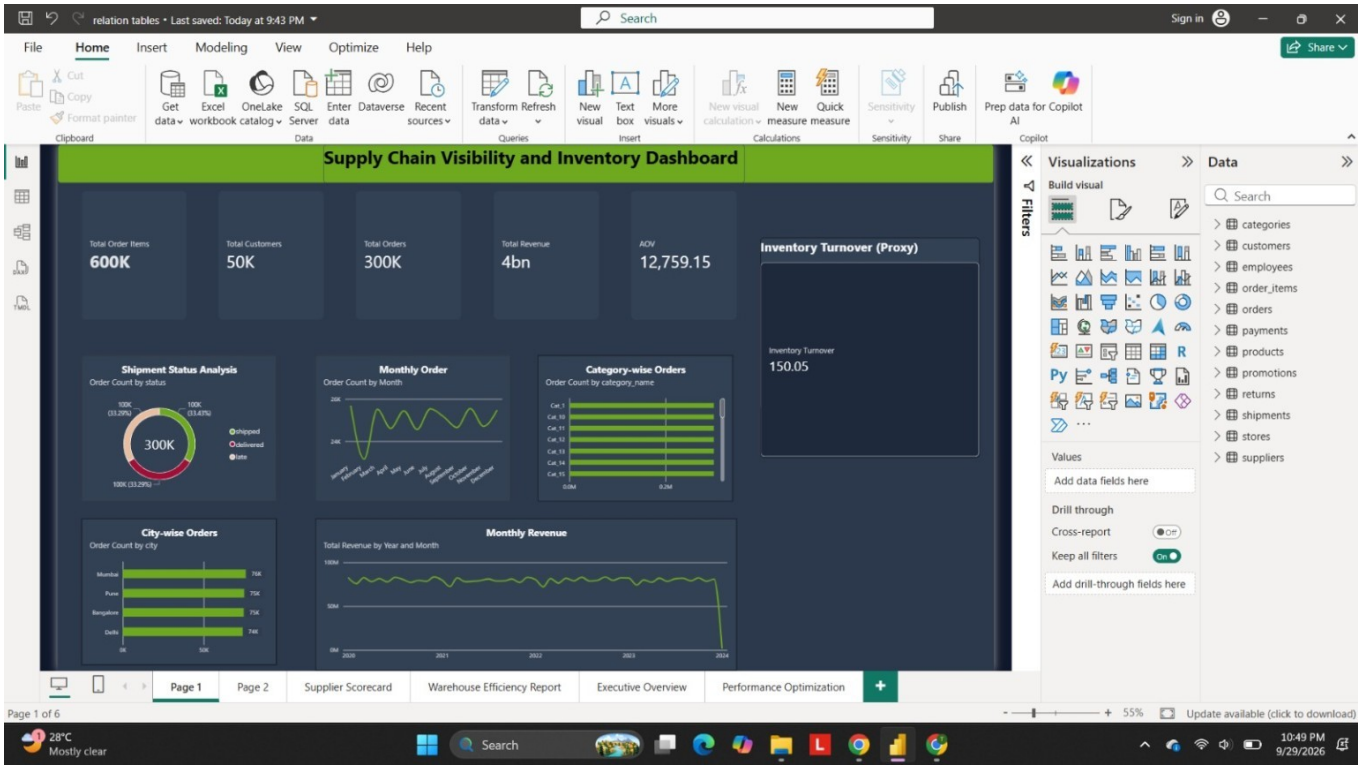


Figure 1: Milestone 1 Dashboard – KPI Foundation

MILESTONE – 2

Inventory & Delivery Analytics

Milestone 2 focuses on inventory and delivery visibility. The supplied screenshot shows inventory KPIs, inventory by product category, inventory by city, inventory trend over time, on-time versus delayed deliveries, delivery performance by city, orders by city and category, delivery trend over time, and an inventory-details table.

Milestone 2 Deliverables

- Develop inventory performance reporting.
- Create delivery performance dashboards.
- Provide regional and product/category analysis through the available report fields.
- Add time-based trend views for inventory and delivery activity.

5.1 Milestone 2 Introduction

The objective of Milestone 2 is to move from the KPI foundation to operational analysis. The supplied dashboard provides a focused view of inventory levels, inventory turnover, product/category distribution, city-level inventory, delivery status, and delivery trends.

5.2 Inventory Performance Analysis

Visible inventory KPI	Value shown
Total Inventory	2M
Inventory Turnover	150.05
Total Products	10K
Slow-Moving Products	0

- Inventory by Product Category is presented as a category-level quantity view.
- Inventory by City is presented as a city-level quantity comparison.
- Inventory Trend Over Time presents quantity by year.
- Inventory Details Table provides product, category, city, quantity, and Inventory Turnover fields in the visible report.

5.3 Delivery Performance Analysis

The supplied dashboard includes an “On-Time vs Delayed Deliveries” visual, a “Delivery Performance by City” visual, and a “Delivery Trend Over Time” visual. Shipment and delivery status categories visible in the project include shipped, delivered, and late.

The report also shows a shipment-status distribution using the same status categories. The exact calculation behind on-time delivery and the underlying DAX expression are not visible in the screenshot and are therefore recorded as: “To be updated from the Power BI report.”

5.4 Regional and Product-Level Analysis

The report provides city-level views for inventory and delivery performance and category/product-level views for inventory. The visible cities include Bangalore, Delhi, Mumbai, and Pune. Product/category details are also presented in the Inventory Details Table and category visuals.

Exact drill-down hierarchy settings and interaction behavior are not visible in the supplied screenshot. Those settings are: “To be updated from the Power BI report.”

5.5 Trend and Variance Analysis

The report contains Inventory Trend Over Time and Delivery Trend Over Time visuals. These provide time-based views of quantity and delivery activity. No additional variance formula is inferred from the screenshots.

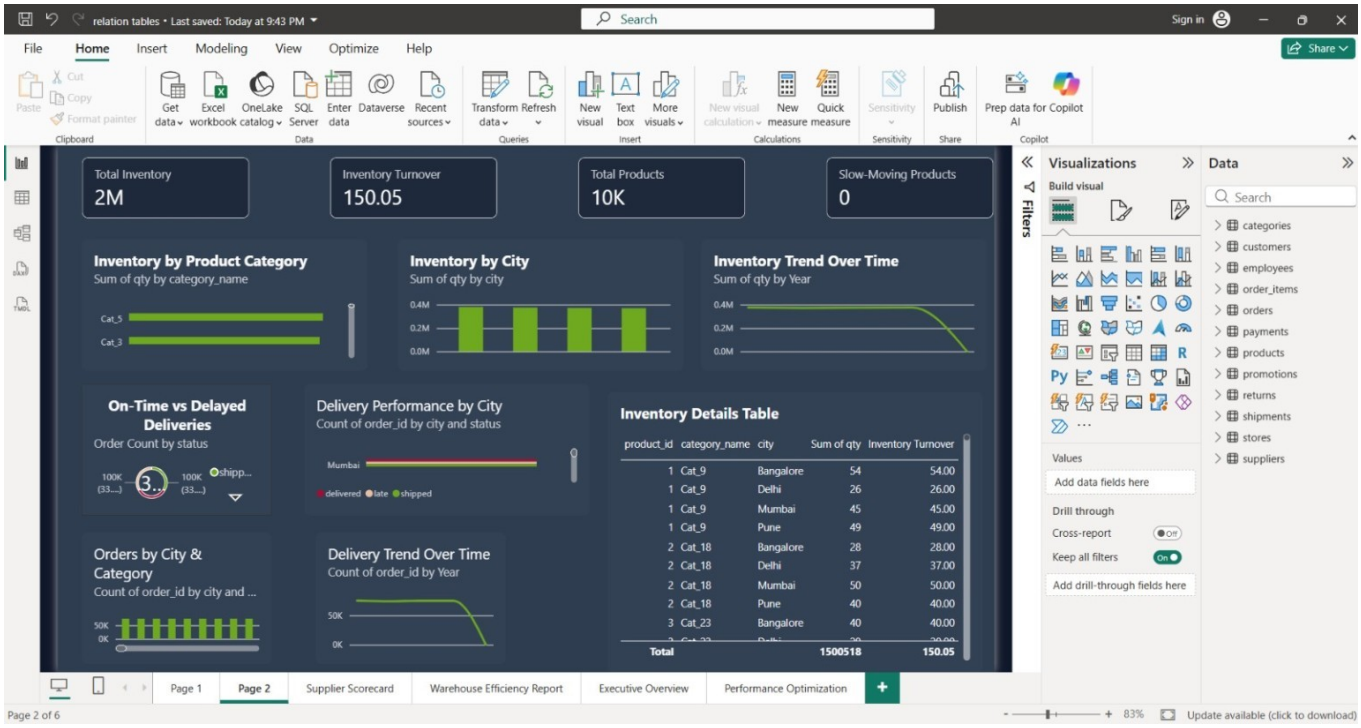


Figure 2: Milestone 2 – Inventory & Delivery Analytics

MILESTONE – 3

Supplier & Transportation Analytics

Milestone 3 documents the supplier-focused reporting visible in the supplied Power BI project and records the transportation-related information only where it is actually visible in the provided dashboard screenshot.

Milestone 3 Deliverables

- Build supplier scorecard reporting.
- Develop supplier comparison visuals.
- Document shipment and transportation information visible in the report.
- Use KPI-style visuals for supplier and shipment monitoring.

6.1 Milestone 3 Introduction

The objective of Milestone 3 is to provide supplier-level visibility and connect supplier information with shipment and inventory-related reporting. The supplied Supplier Scorecard page includes supplier-level cards, shipment KPIs, shipment-status distribution, products by supplier country, supplier-wise inventory value, and inventory value by supplier country.

6.2 Supplier Analysis

The Supplier Scorecard screenshot shows supplier-related KPI cards on the left, including Total Products, Total Inventory, and Inventory Value for selected supplier entries. The page also contains Total Shipments, Delivered Shipments, and Late Shipments KPI cards.

The screenshot includes a “Products by Supplier Country” bar chart for USA, China, and India; a “Supplier-wise Inventory Value” visual; and an “Inventory Value by Supplier Country” chart. A country filter is also visible with China, India, and USA selections.

Visible item	Value shown in supplied screenshot
Total Shipments	300K
Delivered Shipments	99.862K
Late Shipments	100K
Supplier countries visible	USA, China, India

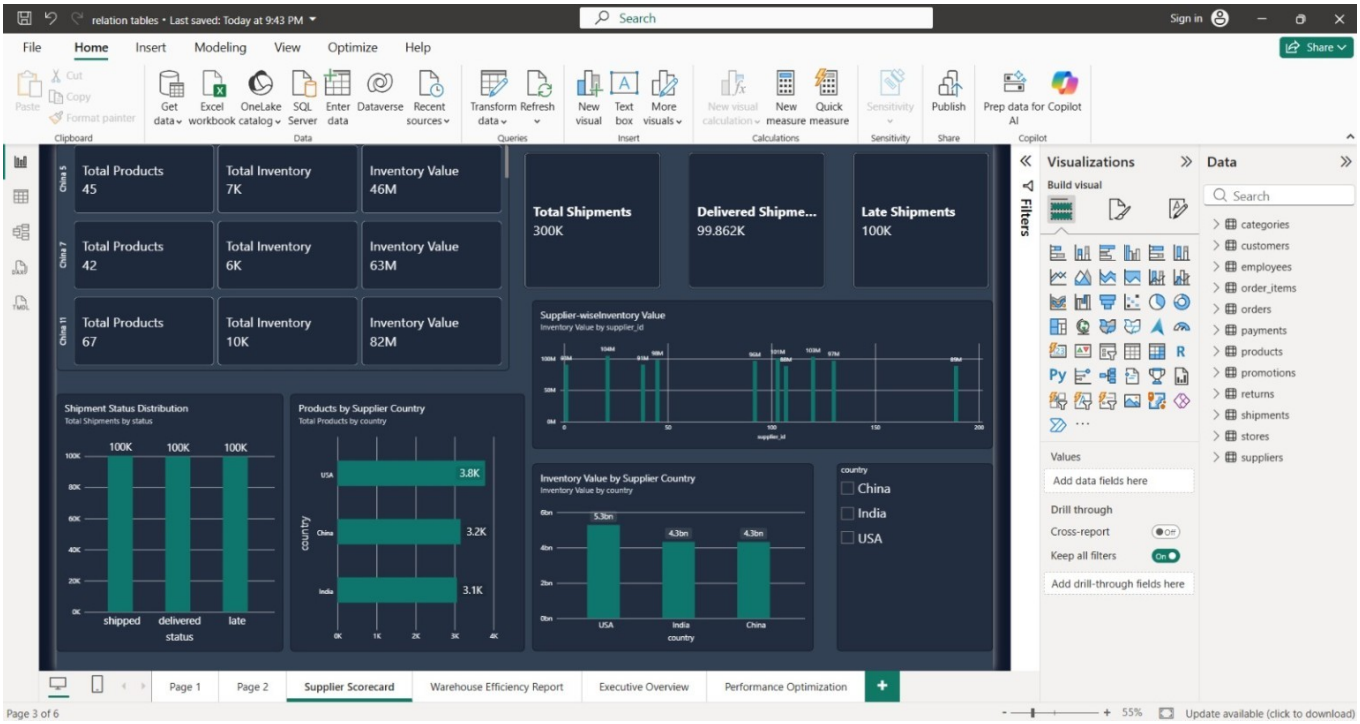


Figure 3: Supplier Performance Scorecard

6.3 Transportation Analysis

The supplied screenshot provides shipment-related indicators and shipment-status distribution, which are relevant to transportation activity. It does not visibly provide a dedicated route-cost or carrier-cost table. Therefore, route analysis, carrier analysis, and transportation cost analysis are not inferred.

- Transportation information visible: shipment activity and shipment status.
- Shipment information visible: Total Shipments, Delivered Shipments, and Late Shipments.
- Route analysis: To be updated from the Power BI report.
- Carrier analysis: To be updated from the Power BI report.
- Transportation cost analysis: To be updated from the Power BI report.

6.4 KPI and DAX Measures

The screenshot confirms the presence of KPI-style measures for shipment activity and supplier-related reporting. Exact DAX formulas and measure definitions are not visible in the supplied screenshot and are therefore: “To be updated from the Power BI report.”

6.5 Transportation Dashboard Screenshots

The supplied screenshot is used as the real Supplier Scorecard evidence for Milestone 3. No separate route/carrier dashboard screenshot was supplied.

Route and Carrier Performance Analysis: To be updated from the Power BI report.

MILESTONE – 4

Warehouse Analytics & Final Dashboard

Milestone 4 documents the Warehouse Efficiency Report, Executive Overview, and Performance Optimization views supplied as the final three screenshots. These pages show the report's operational, executive, and optimization-oriented reporting layers.

Milestone 4 Deliverables

- Develop warehouse efficiency reporting.
- Create an executive overview dashboard.
- Document dashboard performance and layout optimization work visible in the project.
- Record final testing and deployment information only where supported by the supplied report.

7.1 Warehouse Efficiency Report

The Warehouse Efficiency Report screenshot contains slicers for city, category_name, order_date, and status. It also includes KPI cards for Total Orders, Total Shipped Quantity, and On-Time Delivery, together with order/category, shipment-status, city, monthly-order, and detail-table visuals.

Visual / component	Fields or information visibly represented	Purpose
City slicer	city	Filter warehouse/order analysis by city.
Category slicer	category_name	Filter analysis by category.
Date slicer	order_date	Filter the report by order-date range.
Status slicer	status	Filter shipment/order status.
Total Orders	Order count	Show total order activity under the current filter context.
Shipment Count by status	status	Compare shipped, delivered, and late shipment counts.
Total Orders by city	city	Compare order activity across cities.
Total Orders by Month	Month	Show monthly order trend.
Detail table	order_id, shipment_id, status, city, qty	Provide row-level operational detail.

The screenshot also shows an On-Time Delivery KPI with the visible value 7.01E-5. The underlying formula is not visible and is not inferred.

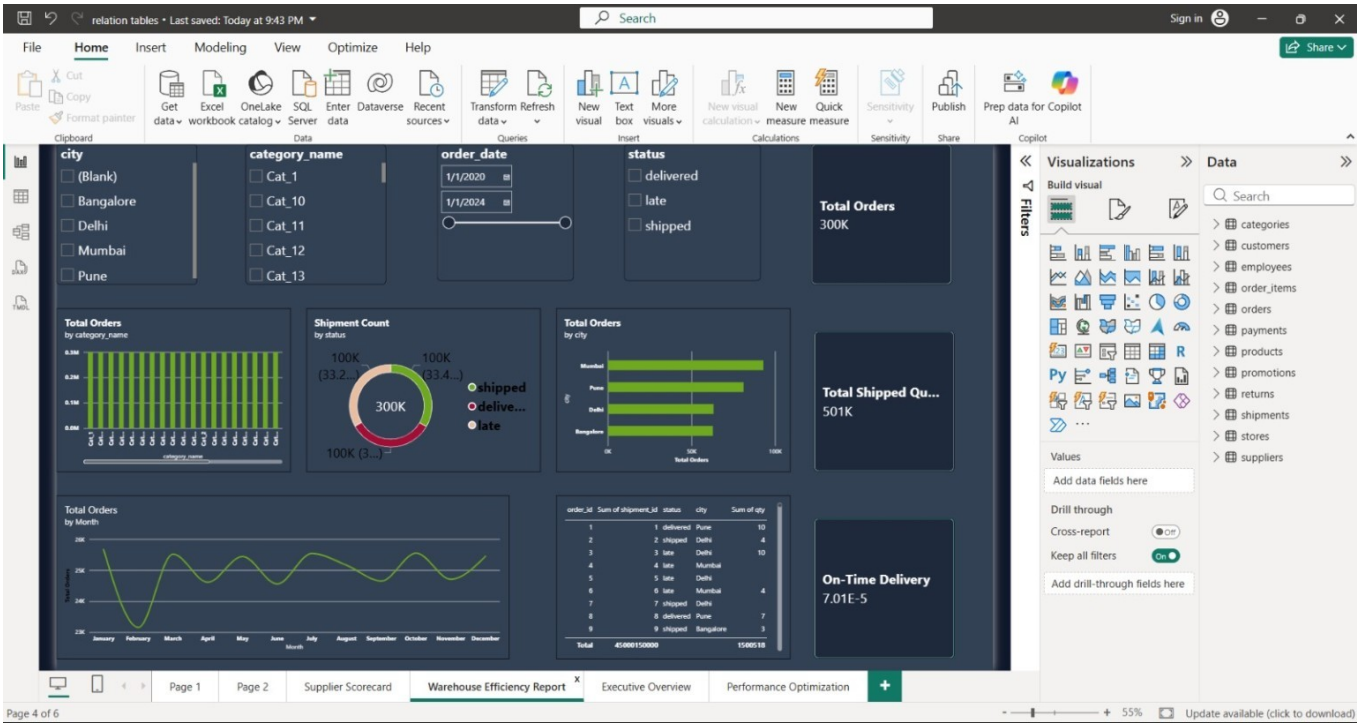


Figure 4: Warehouse Efficiency Report Dashboard

7.2 Executive Overview Dashboard

The Executive Overview page provides a compact management view. The supplied screenshot shows KPI cards for Total Revenue, Total Orders, Total Customers, AOV, and Inventory Turnover. It also contains City-wise Revenue Performance, Total Revenue by category_name, and an order_date slicer.

Executive KPI	Exact visible value
Total Revenue	4bn
Total Orders	300K
Total Customers	50K
AOV	12.76K
Inventory Turnover	150.05

The city-wise revenue chart compares Bangalore, Delhi, Mumbai, and Pune. The category-wise revenue visual displays revenue by category_name. The date slicer visibly spans 1/1/2020 to 1/1/2024 in the supplied screenshot.

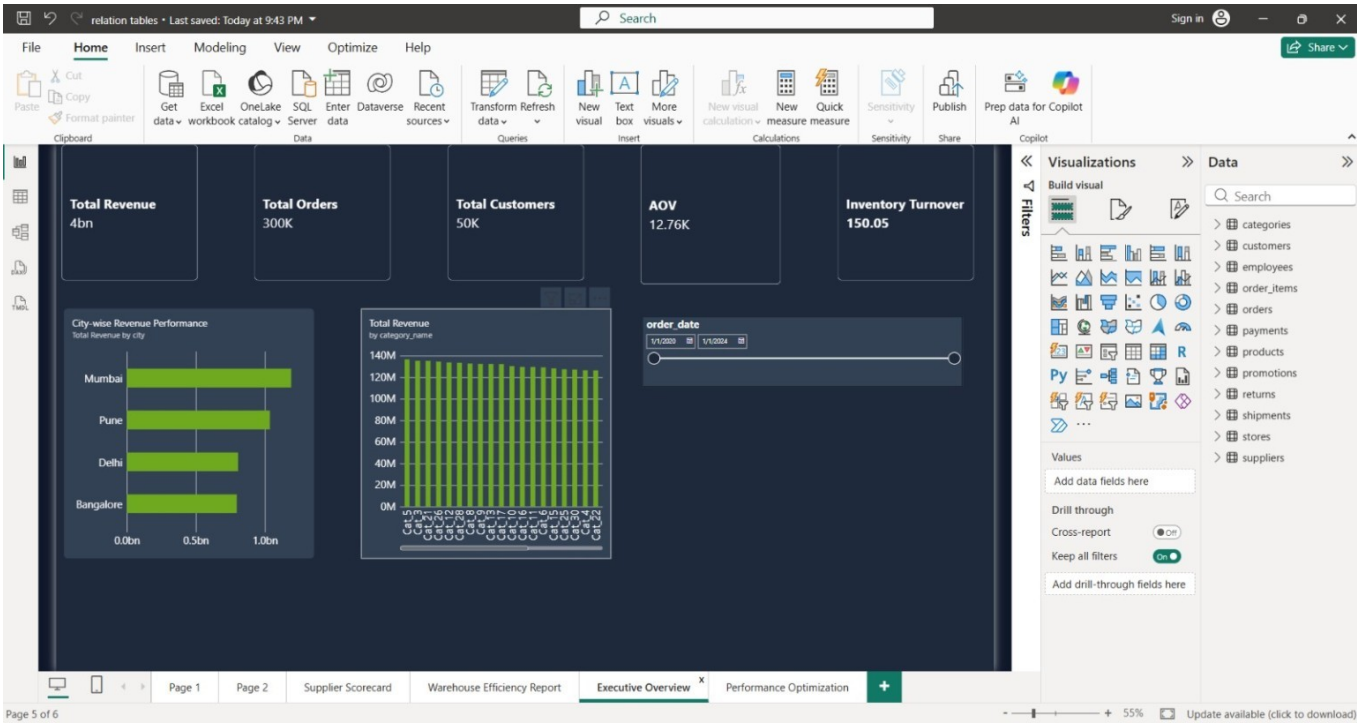


Figure 5: Executive Overview Dashboard

7.3 Dashboard Performance Optimization

The supplied Performance Optimization screenshot shows a compact summary table containing city, total revenue, total orders, total customers, AOV, and Inventory Turnover. The visible table provides a consolidated city-level view of these measures.

- Removing unnecessary visuals: optimization work is not independently evidenced by a before/after comparison in the supplied screenshot; detailed action record is to be updated from the Power BI report.
- Removing unnecessary fields: To be updated from the Power BI report.
- Optimizing visual interactions: To be updated from the Power BI report.
- Reducing unnecessary calculations: To be updated from the Power BI report.
- Using appropriate measures: KPI measures are visibly used across the report.
- Consistent formatting: the supplied pages use a consistent dark dashboard theme and repeated visual styling.
- Clean dashboard layout: the pages use aligned cards, charts, slicers, and tables.
- Slicer/filter optimization: slicers are visibly used for city, category, status, date, and country on the supplied pages.
- Improved readability and navigation: report pages are separated into operational and executive views.

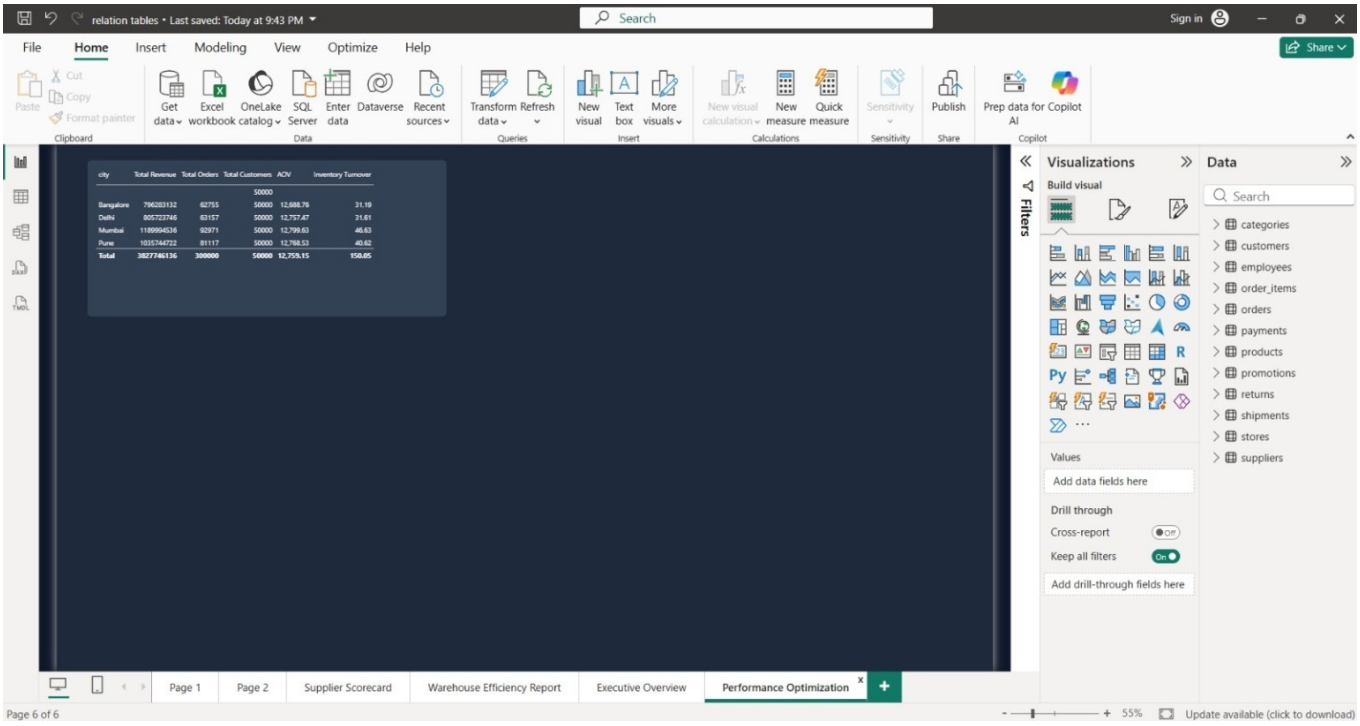


Figure 6: Dashboard Performance Optimization

7.4 Final Testing and Documentation

The supplied screenshots provide visual evidence of multiple report pages and their KPI cards, charts, slicers, and tables. They allow documentation of visible layout and filter components. A detailed test log covering every filter combination, relationship, measure, navigation action, and validation step is not included in the supplied material; therefore, those items are recorded as "To be updated from the Power BI report."

Testing area	Evidence available in supplied material
Visual verification	Visible dashboard screenshots provided.
Filter/slicer verification	Slicers are visible on the Warehouse Efficiency Report, Supplier Scorecard, and other pages.
KPI verification	KPI values are visible on supplied screenshots.
Relationship verification	To be updated from the Power BI report.
Dashboard navigation	Six report pages are visible across the supplied screenshots.
Formatting consistency	Dark-theme dashboard styling is visible across supplied pages.

7.5 Deployment

Deployment or publishing details are not shown in the supplied screenshots. Therefore: "To be updated from the Power BI report."

8. KEY INSIGHTS

The following observations are factual descriptions of what is visible in the supplied Power BI screenshots. They do not introduce additional business results or assumptions.

- The main dashboard reports 600K Total Order Items, 50K Total Customers, 300K Total Orders, 4bn Total Revenue, AOV of 12,759.15, and Inventory Turnover (Proxy) of 150.05.
- The inventory dashboard reports Total Inventory of 2M, Inventory Turnover of 150.05, Total Products of 10K, and Slow-Moving Products of 0.
- Shipment-status reporting uses shipped, delivered, and late categories across the supplied dashboards.
- City-level analysis is available for Bangalore, Delhi, Mumbai, and Pune in the supplied report views.
- Supplier reporting includes supplier-level inventory/product cards, shipment KPIs, supplier-country comparisons, and a country slicer.
- The Warehouse Efficiency Report includes city, category, date, and status slicers and provides order, shipment-status, monthly-order, and detail-table views.
- The Executive Overview consolidates Total Revenue, Total Orders, Total Customers, AOV, and Inventory Turnover in one page.
- The Performance Optimization page provides a compact city-level table combining revenue, orders, customers, AOV, and inventory turnover.

9. REFERENCES

- Microsoft Power BI Documentation — general reference for Power BI Desktop, data modelling, Power Query, DAX, and visualizations.

No additional external references were used as a basis for the factual project content in this document. Project-specific statements are based on the supplied Power BI screenshots and the information provided with the project request.